Linkers

A linker or link editor is program that takes the objects generated by the compiler and assembles them into a single executable program or a library. The result of compiling a source file is an object file. Compiling a project can result in a number of object files. The object files themselves are incomplete, as some of the internal variable and function reference to an external library are still unresolved.

Linker is responsible for combining these object files and to resolve all the unresolved symbols in the object file(s). These unresolved objects are acquired by the linker from a library, eg. libc or the Standard C Library we mentioned earlier. For embedded systems it is advised that we use a subset of the libc something like newlib. Once all the unresolved objects have been resolved to ones in a given library a linkers work is done. There is still one more step for embedded systems though, memory allocation. The generated code must be provided with memory addresses for loading the code and data in the target processor memory. You'll have to provide the information about memory to a program called the locator. The locator uses this information to assign physical memory addresses to each of the code and data sections. The locator produces an output file that contains a binary image which can be directly loaded into the target ROM. A commonly used linker/locater for embedded systems is GNU ld.

Build Systems

Building a software for just one target system is fairly easy to manage. You just need a proper project structure in place and properly configured compiler and linker/loader toolchain. If you plan on developing and deploying to multiple systems at the same time, you'll be better of with a build system. Build systems like the few mentioned below provide an easy way to manage the build process for your programs. These systems only require some configuration for your target platform, it will manage the rest of the nitty-gritties by itself.

Buildroot *http://www.buildroot.net/*
Buildroot-ng *http://wiki.openwrt.org/*
crosstool-NG *http://www.crosstool-ng.org/*
PTXdist *http://www.ptxdist.org/*
OpenEmbedded *http://www.openembedded.org/*
OE-lite *http://oe-lite.org/*